BTS-VII(R/S)-11-23-1957 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

B

B.Tech. Degree VII Semester Regular/Supplementary Examination November 2023

ME 19-205-0703 MACHINE DESIGN II

(2019 *Scheme*)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the customers need, formulate the problem and draw the design specifications.

CO2: Understand component behavior subjected to loads and identify the failure criteria.

CO3: Apply the concepts of stress analysis, theories of failure and material science to analyze, design and/or select commonly used machine components.

CO4: Analyze the stresses and strains induced in a machine element.

CO5: Design machine components using theories of failure.

CO6: Get basic knowledge of the preparation of working drawings by including the manufacturing details like tolerance, surface finish etc.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

(Use of Design Data Handbook is permitted)

(Data not given may be suitably assumed. All such assumptions must be clearly stated)

PART A

(Answer ***ALL*** questions)

		$(8 \times 3 = 24)$	Marks	BL	CO	PO
I.	(a)	Explain the theory which is suitable for worn out clutches.	3	L2	2	1
	(b)	Explain the function of tension member in V-Belts.	3	L2	2	1
	(c)	What are the factors considered for deciding the type of gear for a particular application?	3	L1	1	1
	(d)	What are the assumptions considered in Lewis equation?	3	L1	3	1
	(e)	Explain the theory of hydrodynamic lubrication with neat sketch.	3	L1	3	1
	(f)	Differentiate between median life and rating life of a bearing.	3	L1	4	1
	(g)	Write the design recommendations for a rolled steel beam.	3	L1	5	1
	(h)	Find the type of fit in the following case.(all dimensions are in mm) Size of Hole = $25.00^{+0.013}_{+0.000}$, Size of Shaft = $25.00^{-0.000}_{-0.013}$	3	L1	6	1

PART B

$(4 \times 12 = 48)$

			Marks	BL	CO	PO
II.	(a)	A cone clutch is used to connect an electric motor running at 1440 rpm with a machine which is stationary. The machine is equivalent to a rotor of 150 kg mass and radius of gyration as 250 mm. The machine has to be brought to the full speed of 1440 rpm from stationary condition in 40 s. The semi-cone angle α is 12.5°. The mean radius of the clutch is twice the face width. The coefficient of friction is 0.2 and the normal intensity of pressure between contacting surfaces should not exceed 0.1 N/mm^2. Assuming uniform wear criterion, calculate: (i) the inner and outer diameters; (ii) the face width of friction lining (iii) the force required to engage the clutch.	6	L3	2	1

(P.T.O.)

		Marks	BL	CO	PO
(b)	A centrifugal clutch, transmitting 20 kW at 750 rpm consists of four shoes. The clutch is to be engaged at 500 rpm. The inner radius of the drum is 165 mm. The radius of the center of gravity of the shoes is 140 mm, when the clutch is engaged. The coefficient of friction is 0.3, while the permissible pressure on friction lining is 0.1 N/mm^2. Calculate: (i) The mass of each shoe (ii) The dimensions of friction lining.	6	L3	2	1
	OR				
III. (a)	A 15 kW, 1100 rpm motor drives a line shaft at 200 rpm. The shaft center distance is approximately 600 mm. The motor shaft has a diameter of 50 mm. The starting torque on the motor is 2 times the running torque. The load is applied for 10 hours with moderate shocks. Select a suitable chain drive and calculate the exact length and exact center distance of the chain.	8	L3	3	1
(b)	A 4 ply balata belt is used to transmit power from 75 mm diameter wooden pulley to a 225 mm diameter CI pulley to run at 400 rpm. Find the power capacity of this belt drive. The belt used is 10 mm thick and 90 mm wide. Take the centre distance as 1 m.	4	L3	3	1
IV.	A pair of spur gears consists of a 24 teeth pinion, rotating at 1000 rpm and transmitting power to a 48 teeth gear. The module is 6 mm, while the face width is 60 mm. Both the pinion and gear are made of steel with S_{ut} of 450 N/mm^2. They are heat treated to surface hardness of 250 BHN. The gears are machined to the accuracy of grade 8. Calculate: (i) Beam strength (ii) Wear strength (iii) Dynamic load by Spott's equation (iv) Rated power that the gears can transmit, if the service factor and the factor of safety are 1.5 and 2 respectively.	12	L3	4	1
	OR				
V.	The vertical spindle of a drilling machine is to be driven by a pair of straight bevel gears with 20° teeth. The speed is 4:1. The drill requires a power of 50 kW at 720 rpm. A service factor of 1.35 may be taken, and choose the steel (C30) as material for both gear and pinion. Design the gear pair.	12	L3	4	1
VI.	It is required to design a main bearing of a 4-stroke oil engine to sustain a load of 6 kN over a shaft of diameter 50 mm. The operating speed of the shaft is 1000 rpm and the operating temperature is 50° C. Determine: (i) Dimensions of the bearing. (ii) The viscosity of the oil to be used, hence suggest appropriate oil. (iii) Coefficient of friction. (iv) Heat generated. (v) Heat dissipated. (vi) Heat to be removed by artificial cooling if necessary.	12	L3	5	1
	OR				

BTS-VII(R/S)-11-23-1957

VII.	(a)	A single row deep groove ball bearing is subjected to a radial force of 8000 N and a thrust force of 3000 N. The shaft rotates at 1200 rpm. The expected life of the bearing is 20,000 hr. The minimum acceptable diameter of the shaft is 75 mm. Select a suitable ball bearing	6	L3	5	1
	(b)	Select a suitable ball bearing for a drilling machine spindle of diameter 40 mm rotating at 300 rpm. It is subjected to a radial load of 2000 N and an axial load of 1000 N. It is to work for 45 hours a week for one year.	6	L3	5	1
VIII.	(a)	What are the general design recommendations for a welded frame? Explain.	6	L2	6	1
	(b)	What are the general design recommendations for forgings? Explain.	6	L2	6	1
		OR				
IX.	(a)	What are the general design recommendations for castings? Explain.	6	L2	6	1
	(b)	Write short notes on tolerance and surface finish.	6	L2	6	1

Bloom's Taxonomy Levels
L1 = 25%, L2 = 25%, L3 = 50%.

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